

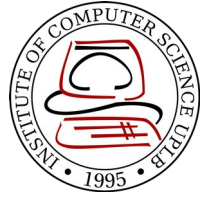
CMSC 137

Data Communications and Networking

Lecture Module
17 – Wireless LAN



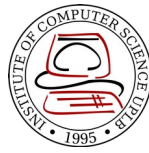
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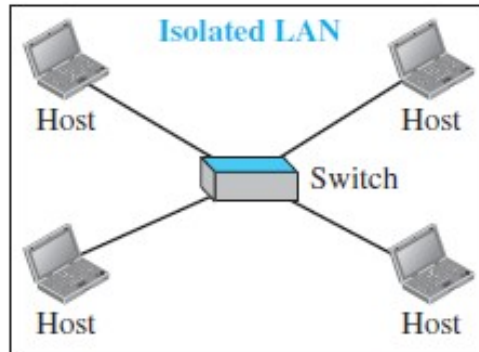
Module 17: Wireless LAN

Rozano S. Maniaol
Lecturer

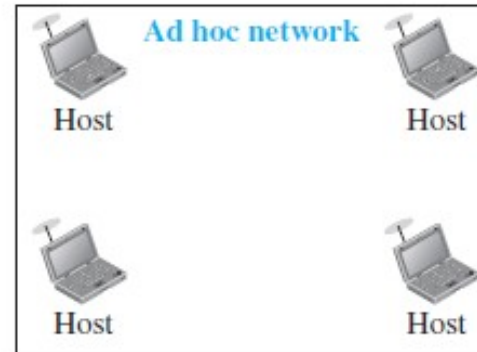
Introduction



- Architectural Comparison
 - **Medium**: Wired LANS use wires, wireless LANs use air
 - **Hosts**: Wired LAN is always connected with fixed link layer address, wireless LAN not physically connected
 - **Isolated LANs**: Wired LAN is a set of host connected via a link layer switch, wireless LAN is a set of hosts that can communicate freely

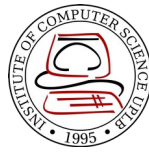


Wired

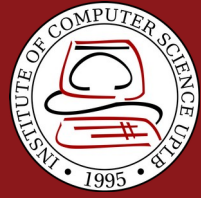


Wireless

Introduction



- Characteristics
 - **Attenuation**: signals decrease rapidly since it is dispersed in all directions
 - **Interference**: may receive from other senders if they are using the same frequency band
 - **Multipath propagation**: may receive more than one signal from the same sender due to reflection on from some obstacles; may receive signal at different phases
 - **Error**: errors and error detection are more serious issues in wireless network



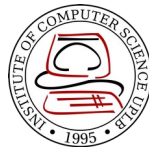
IEEE 802.11

IEEE 802.11



- Specifications for wireless LAN which covers the physical and data link layer (sometimes called wireless Ethernet)
- **Wireless Fidelity** (WiFi) – wireless LAN that is certified by the WiFi Alliance
- Architecture:
 - Defines two kinds of services: basic service set (BSS) and extended service set (ESS)

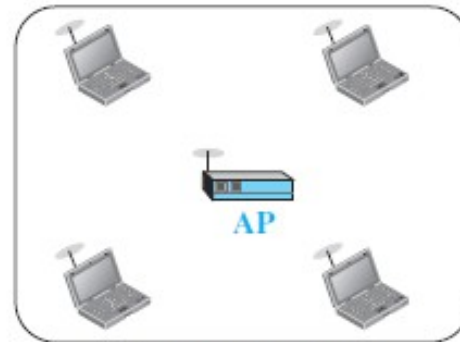
Basic Service Set



- **BSS** – building block of a wireless LAN
 - Made up of stationary or mobile wireless stations and an optional central base station, **access point** (AP)
 - **Ad hoc** – BSS without AP
 - **Infrastructure** – BSS with AP



Ad hoc BSS

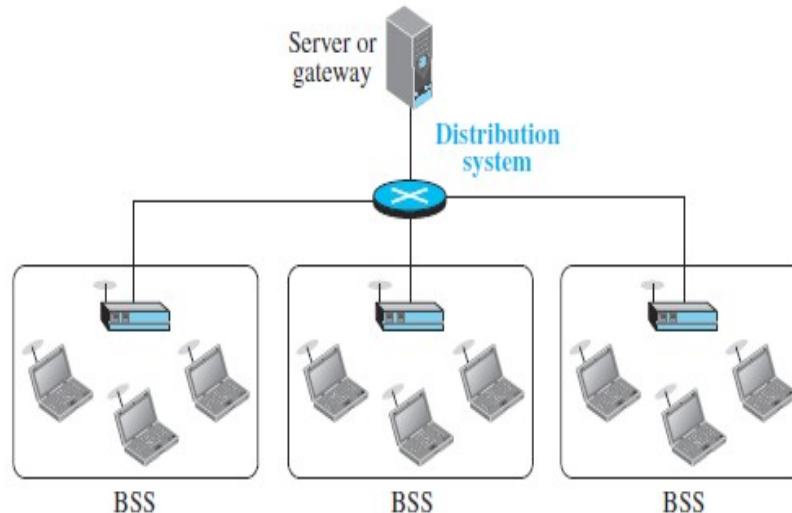


Infrastructure BSS

Extended Service Set



- **ESS** – made up of two or more BSSs with APs; BSS are connected through a **distribution system**, usually a wired LAN
- Mobile or stationary stations

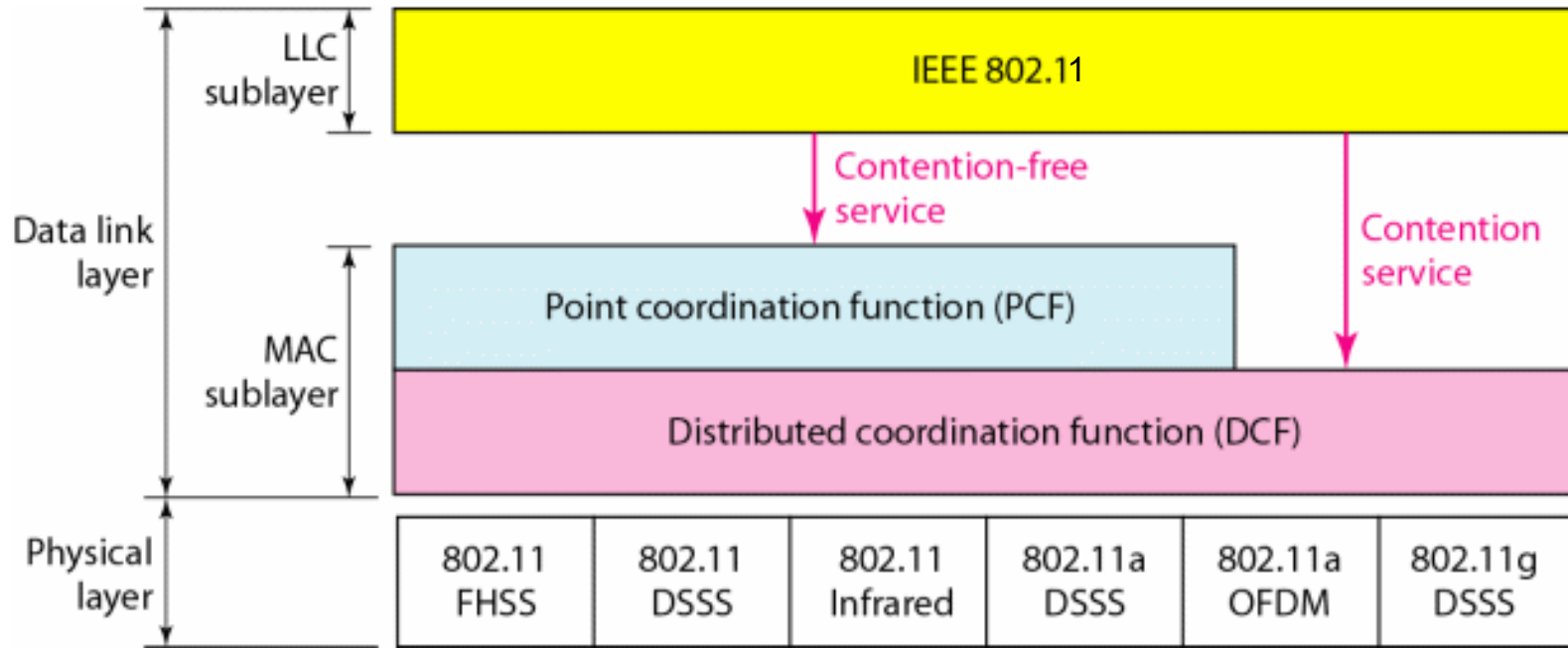


IEEE 802.11

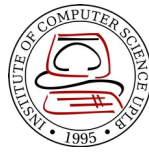


- Station Types:
 - **No-transition** – stationary or moving only inside a BSS
 - **BSS-transition** – station can move from one BSS to another confined inside one ESS
 - **ESS-transition mobility** – station can move from one ESS to another
- **IEEE does not guarantee that communication is continuous**

IEEE 802.11 MAC Layer



Distributed Coordination Function (DCF)



- Uses CSMA/CA
- Cannot use CSMA/CD because:
 - Station must send and receive at the same time; costly
 - Collision may not be detected because of hidden station problem
 - Distance between stations may be great

CSMA/CA Flowchart

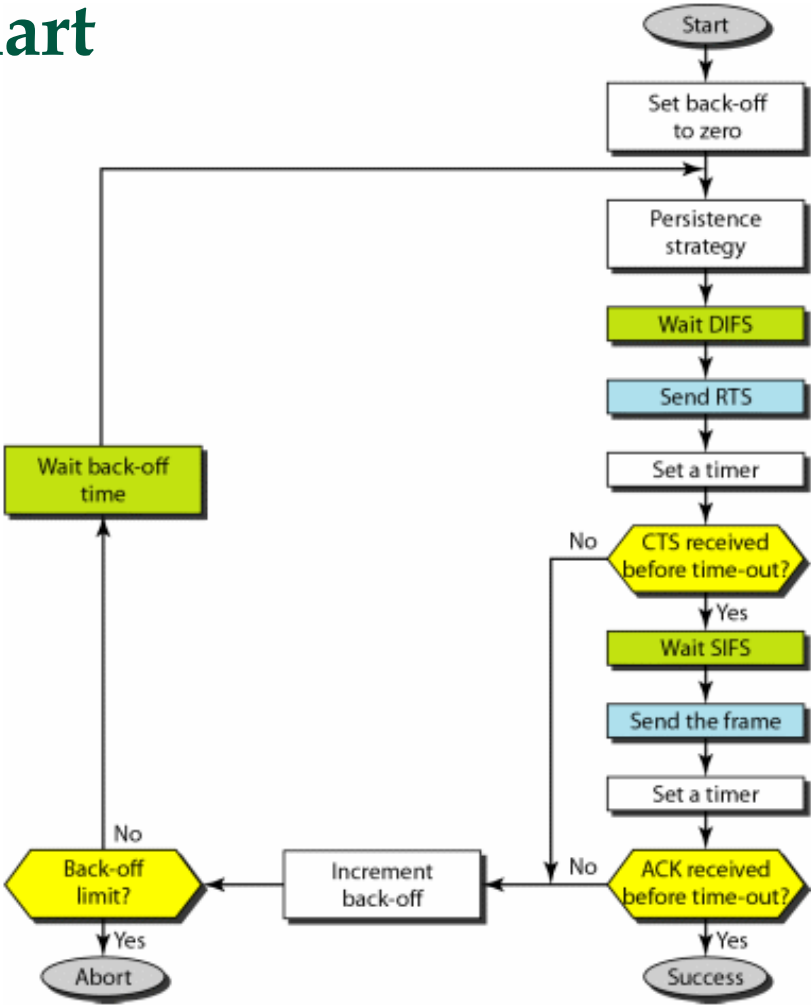
Abbreviations

DIFS – distributed interframe space

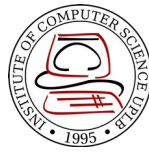
RTS – request to send

SIFS – short interframe space

CTS – clear to send

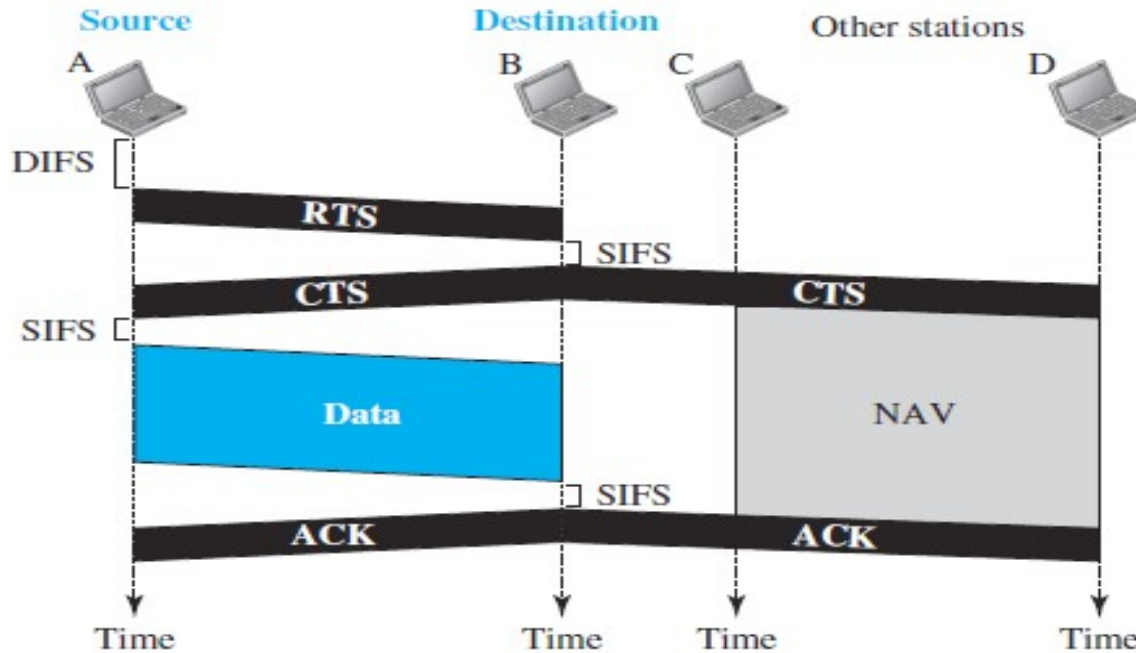


Network Allocation Vector (NAV)



- How other stations defer sending their data if one station acquires access?
 - When a station sends an **RTS** frame, it includes the duration of time it needs to occupy the channel
 - Station affected by the transmission creates a timer called **NAV**
- Each station, before sensing the physical medium to see if it is idle, first checks to see its NAV to see if it has expired
- During handshaking – **no CTS means collision**

CSMA/CA and NAV

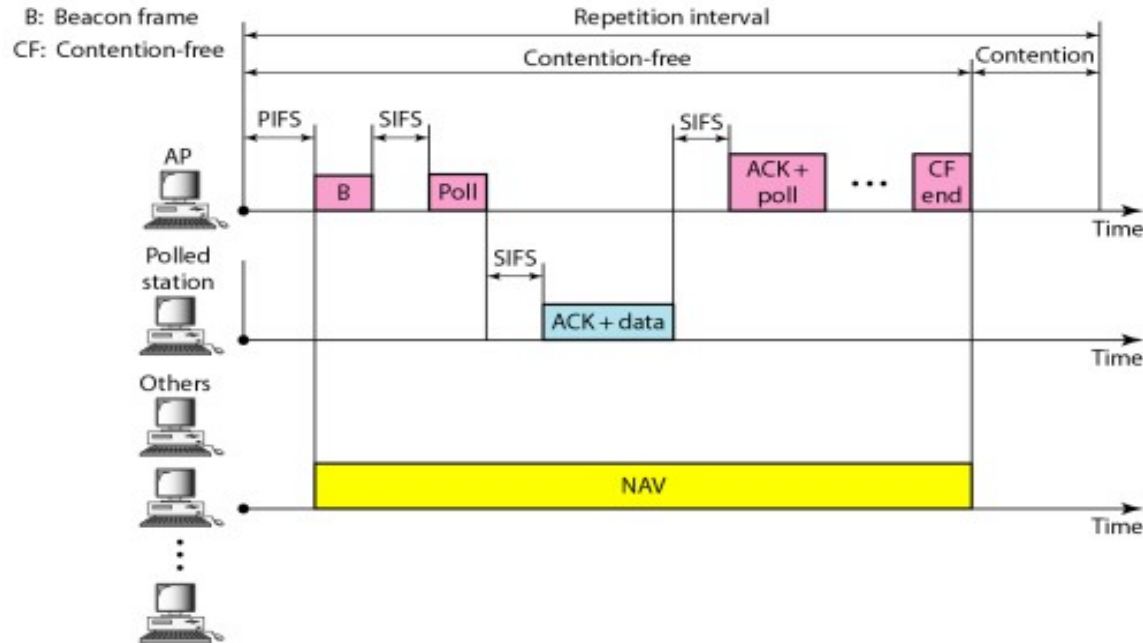


Point Coordination Function (PCF)



- Optional access method for time-sensitive transmission
- Can be implemented in an infrastructure network only
- Centralized, contention-free polling access method
- **PIFS** and **SIFS**, PIFS shorter than DIFS (gives higher priority to PCF over DCF)

Repetition interval - allow DCF stations to access the medium (when PCF is present)

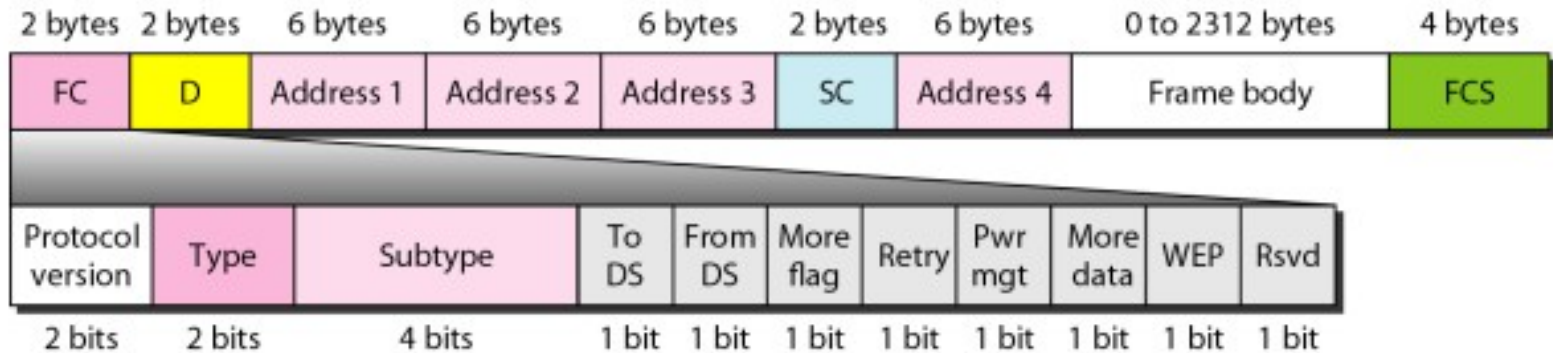


Fragmentation

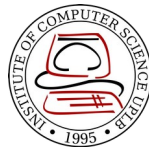


- Wireless environment is very noisy
- Corrupted frames must be re-transmitted
- More efficient to resend small frames than a large one

IEEE 802.11 Frame Format



IEEE 802.11 Frame Format



Frame Control (FC) – defines the type of frame and some control information

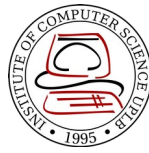
<i>Field</i>	<i>Explanation</i>
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 14.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

IEEE 802.11 Frame Format

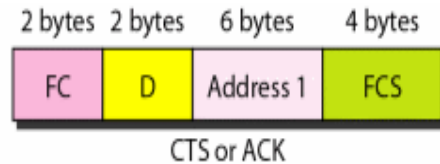
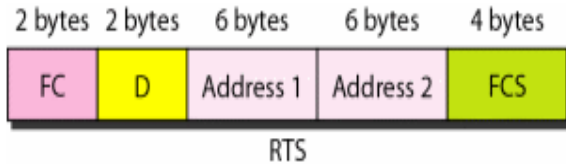


- **D** – duration of the transmission that is used to set the value of NAV
- **Addresses** – four 6-byte addresses (depends on the value of TO DS and FROM DS)
- **Sequence control** – sequence number of the frame to be used in flow control
- **Frame body** – can be between 0 and 2312 bytes; info depends on the type and subtype in FC
- **FCS** - CRC-32

Frame Types

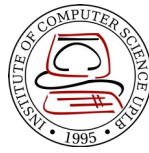


- Management Frames – initial communication
- Data Frames - data
- Control Frames – accessing the channel and acknowledging frames; type field: 01

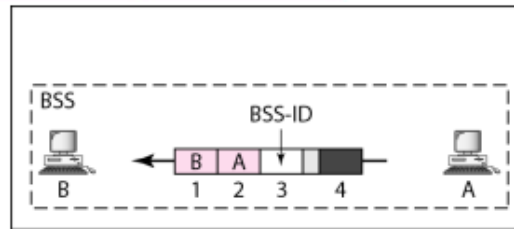


Subtype	Meaning
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

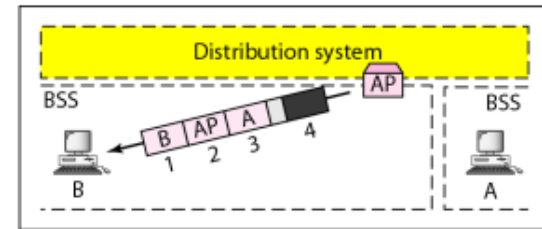
Addressing Mechanism



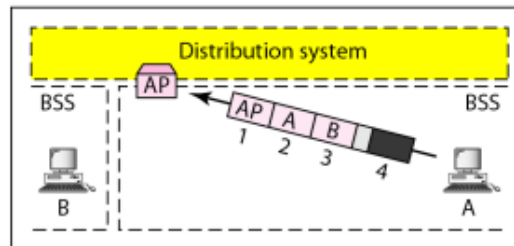
To DS	From DS	Address 1	Address 2	Address 3	Address 4
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source



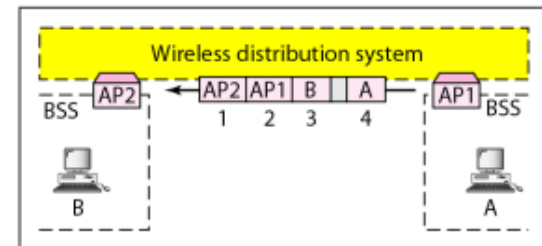
a. Case 1



b. Case 2



c. Case 3

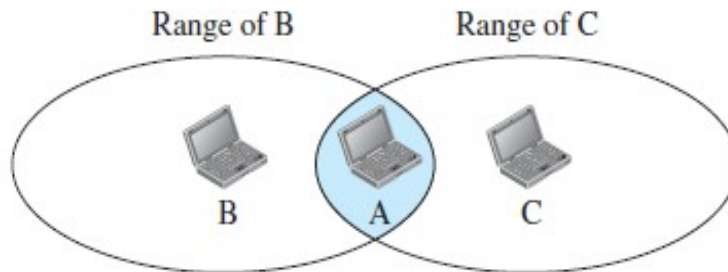


d. Case 4

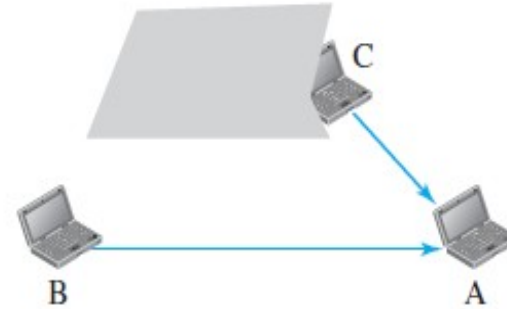
Hidden Station Problem



Can reduce the capacity of the network because of collision

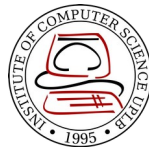


a. Stations B and C are not in each other's range.

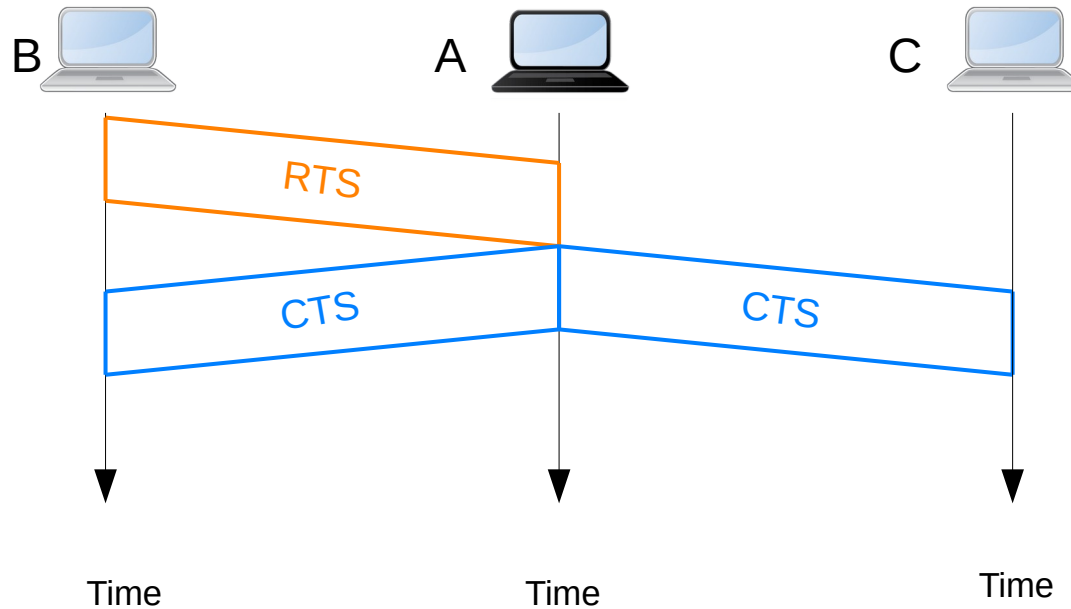


b. Stations B and C are hidden from each other.

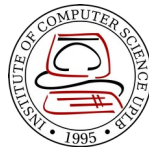
Hidden Station Problem



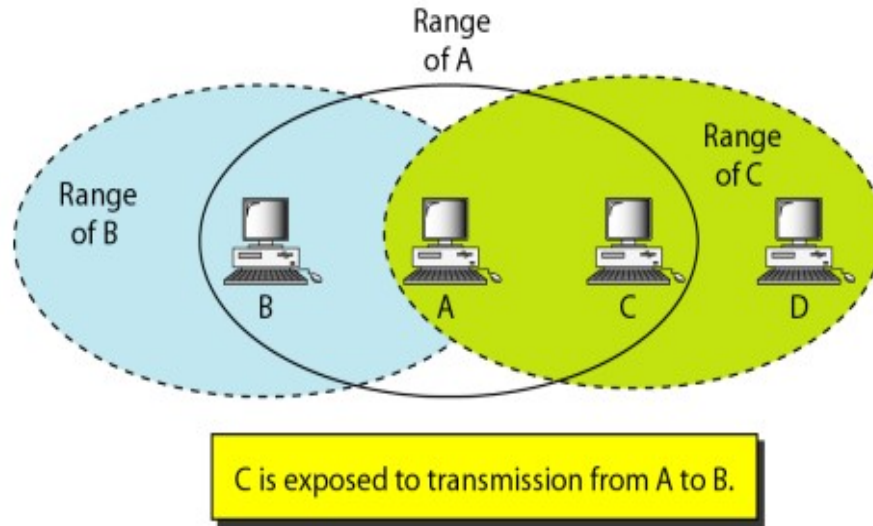
Handshaking to solve HSP using CTS frame



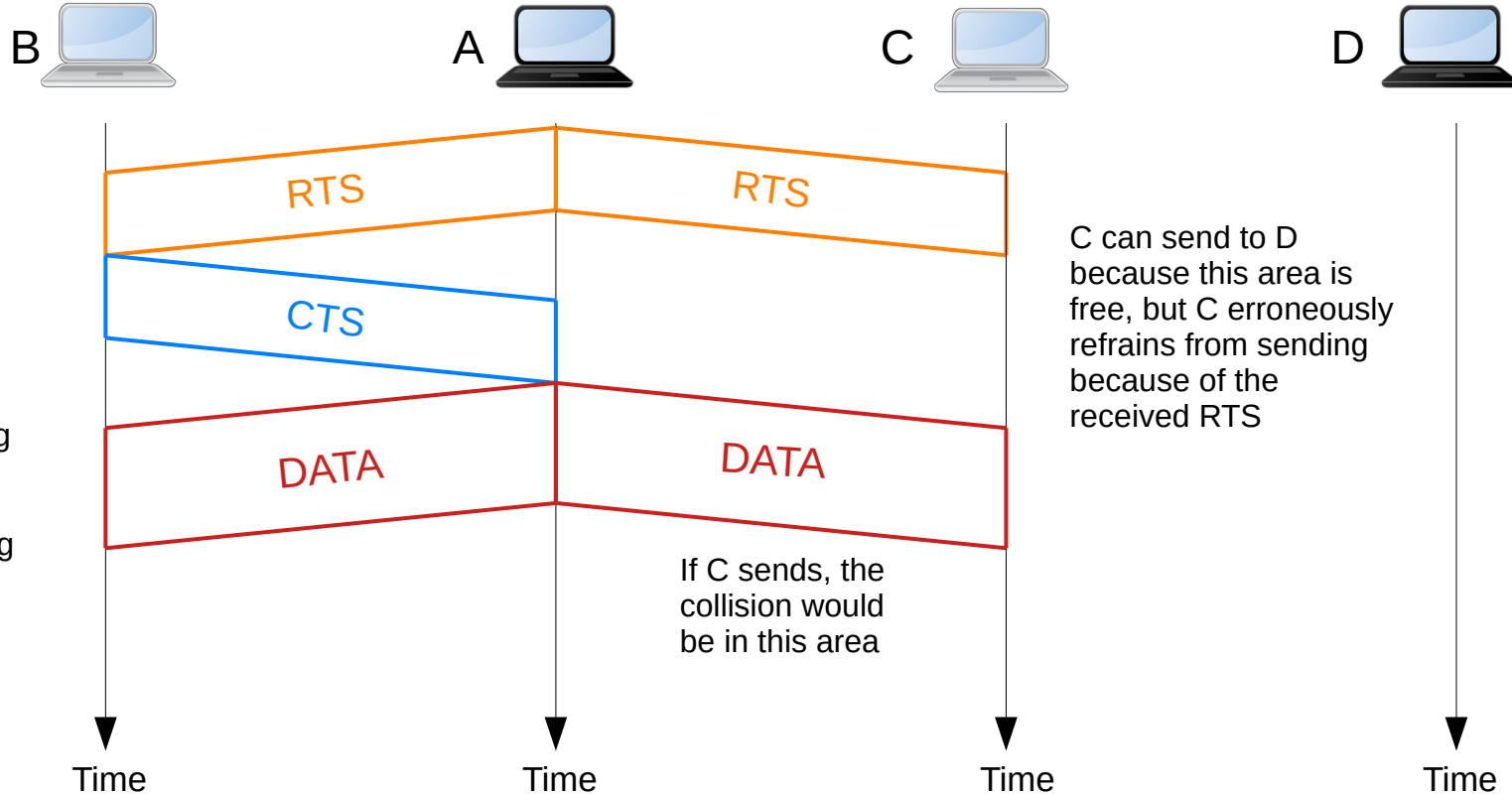
Exposed Station Problem



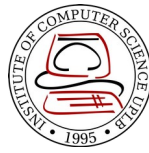
Station refrains from using the channel even when it is available



Exposed Station Problem

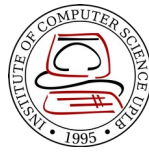


IEEE 802.11 Physical layer

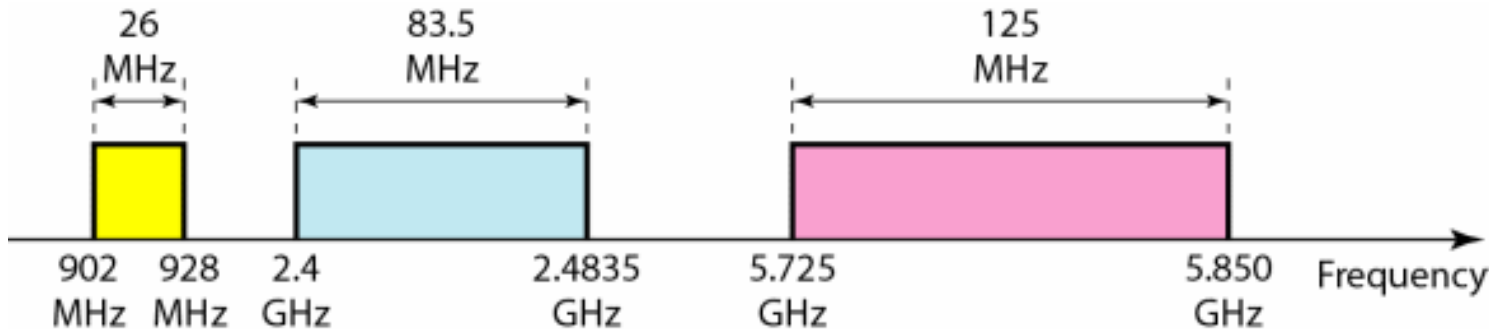


<i>IEEE</i>	<i>Technique</i>	<i>Band</i>	<i>Modulation</i>	<i>Rate (Mbps)</i>
802.11	FHSS	2.400–4.835 GHz	FSK	1 and 2
	DSSS	2.400–4.835 GHz	PSK	1 and 2
	None	Infrared	PPM	1 and 2
802.11a	OFDM	5.725–5.850 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.400–4.835 GHz	PSK	5.5 and 11
802.11g	OFDM	2.400–4.835 GHz	Different	22 and 54
802.11n	OFDM	5.725–5.850 GHz	Different	600

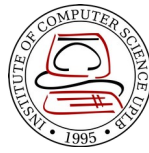
ISM Band



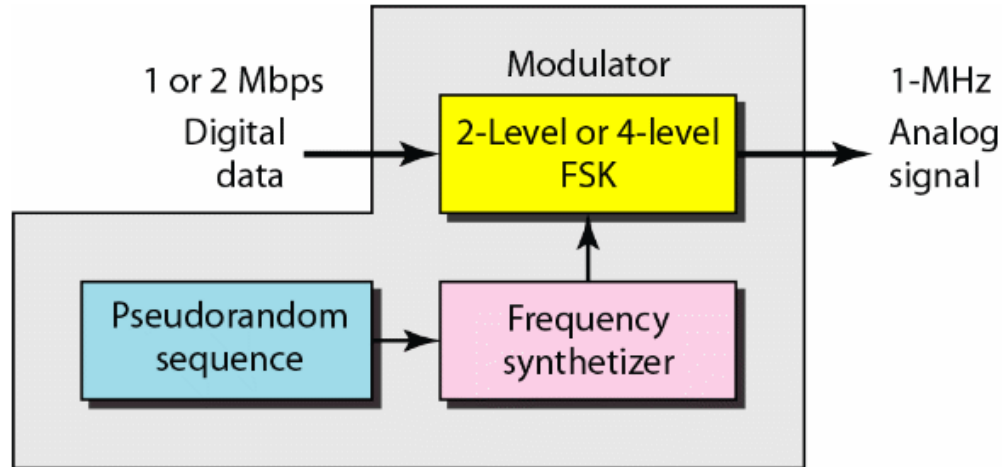
- Internationally reserved portion of the radio frequency bands for Industrial, Scientific, and Medical uses; no need for government licenses
- Opened up for Wireless LANs and mobile communications,



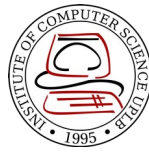
IEEE 802.11 FHSS



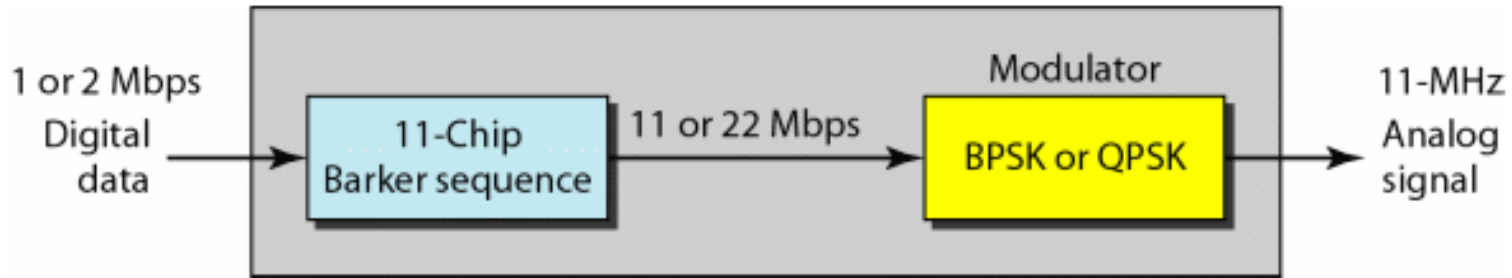
Band is divided into 79 subbands of 1 MHz (and some guard bands)



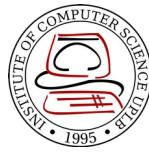
IEEE 802.11 DSSS



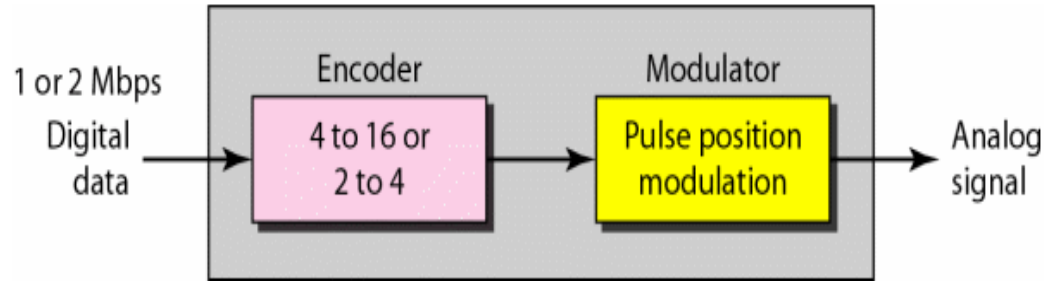
Modulation technique is PSK at 1 Mbaud/s



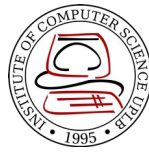
IEEE 802.11 Infrared



- Infrared light in the range of 800 to 950 nm
- Pulse Position Modulation (PPM)
- Mapped sequence are converted to optical signals, presence of light specifies 1, absence of light specifies 0

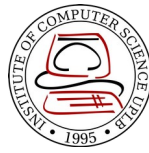


IEEE 802.11a OFDM

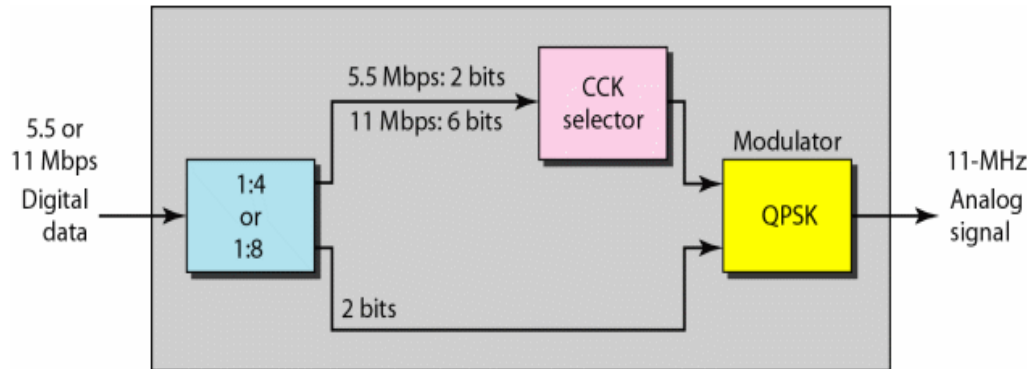


- [Orthogonal frequency-division multiplexing](#) (OFDM) - same as FDM except all sub bands are used by one source at a given time
- 5.725 – 5.850 GHz ISM band
- Band is divided into 52 subbands with 48 groups of bits at a time and 4 subbands for control information
- Uses PSK (18 Mbps) and QAM (54 Mbps)

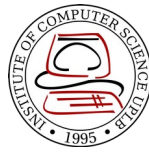
IEEE 802.11b DSSS



- High-rate direct sequence spread spectrum (HR-DSSS)
- 2.400 – 4.835 GHz ISM
- Uses complementary code keying (CCK)
- Encode 4 or 8 bits to one CCK symbol

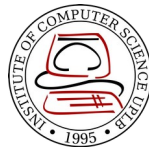


IEEE 802.11g



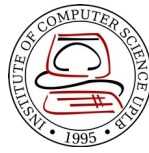
- Defines forward correction and OFDM using the 2.400 – 4.835 GHz ISM band
- 22- or 54-Mbps data rate
- Backward compatible with 802.11b but modulation technique is OFDM

IEEE 802.11n

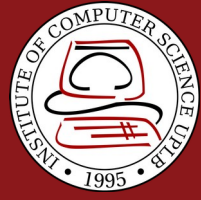


- Upgrade to 802.11 project called 802.11n (next generation of wireless LAN)
- Goal is to increase throughput of wireless LAN (some have reached up to 600 Mbps data rate)
- Eliminated some unnecessary overhead
 - Uses **multiple-input multiple-output antenna** (MIMO)
 - If it can send multiple output signals and receive multiple signals, it is in better position to eliminate noise
- Relabeled as WiFi 4

Latest standards

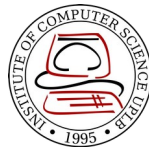


- **802.11ac** – relabeled as WiFi 5
 - Frequency band: 5.8 Ghz ISM band; Max data rate: 6.93 Gbps
 - Implements **Multi-User MIMO** (MU MIMO) which enables simultaneous transmission of different data frames to different clients
- **802.11ax** – relabeled as WiFi 6
 - Frequency bands: 2.4 Ghz and 5 Ghz, 6Ghz for WiFi 6E
 - Multiuser technology: OFDMA + MU MIMO
 - Maxdata Rate: 600.4 Mbps (80 Mhz channel & 1 spatial stream), 9607.8 Mbps (160 Mhz channel & 8 spatial streams)



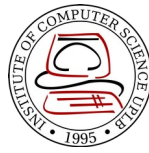
Bluetooth

Bluetooth



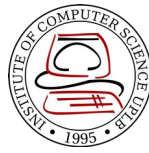
- Wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, etc.
- An ad hoc network - [piconet](#)
- Network cannot be large
- Started as a project by the Ericsson Company
- From the word “Blaatand”

Bluetooth

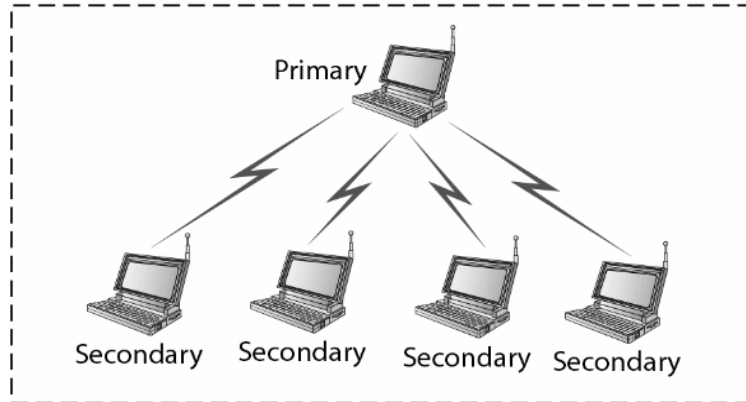


- Piconet
 - May have up to 8 nodes; one primary others secondary
- Scatternet
 - Combination of Piconets
 - A secondary station in one piconet can be the primary in another
 - A station can be a member of two piconets
- Current data rate of 1Mbps with 2.4 GHz bandwidth – can cause interference with 802.11b

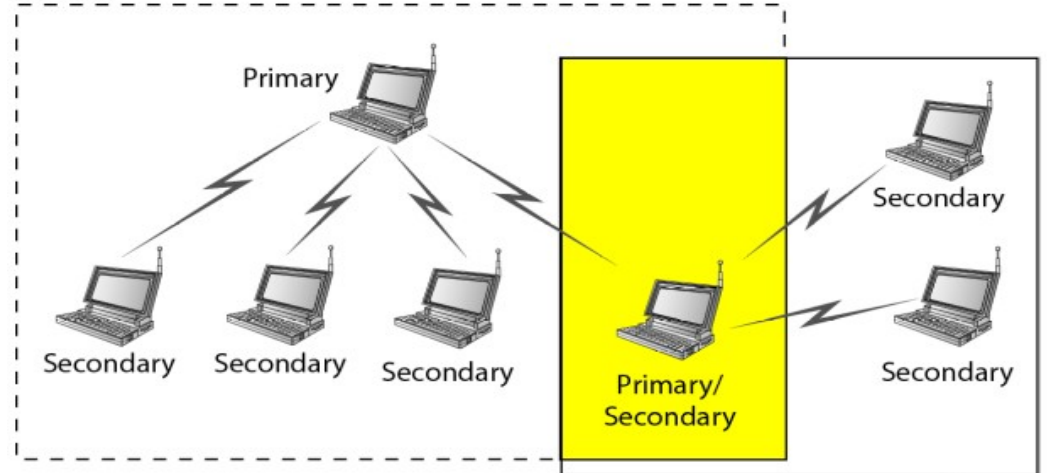
Bluetooth



Piconet

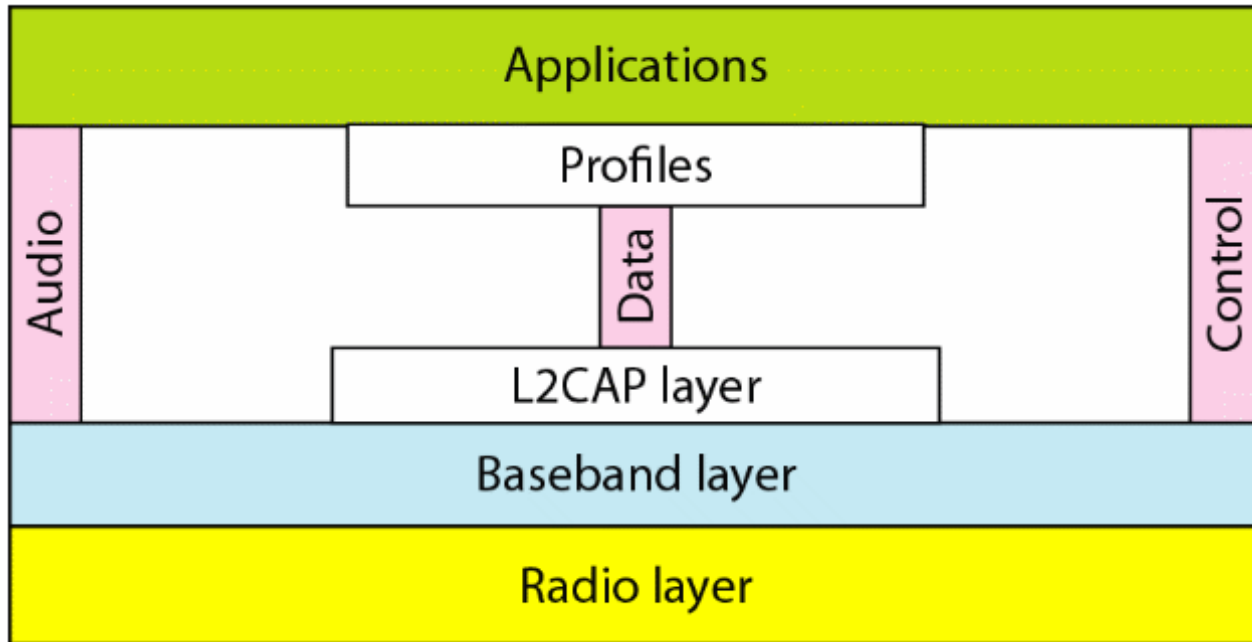


Piconet



Piconet

Bluetooth Layers



Radio Layer



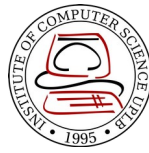
- Equivalent to the physical layer
- Range of 10m
- 2.4-GHz ISM band divided into 79 channels of 1 MHz each
- FHSS – hops 1600 times per second
- Uses Gaussian FSK for modulation

Baseband Layer

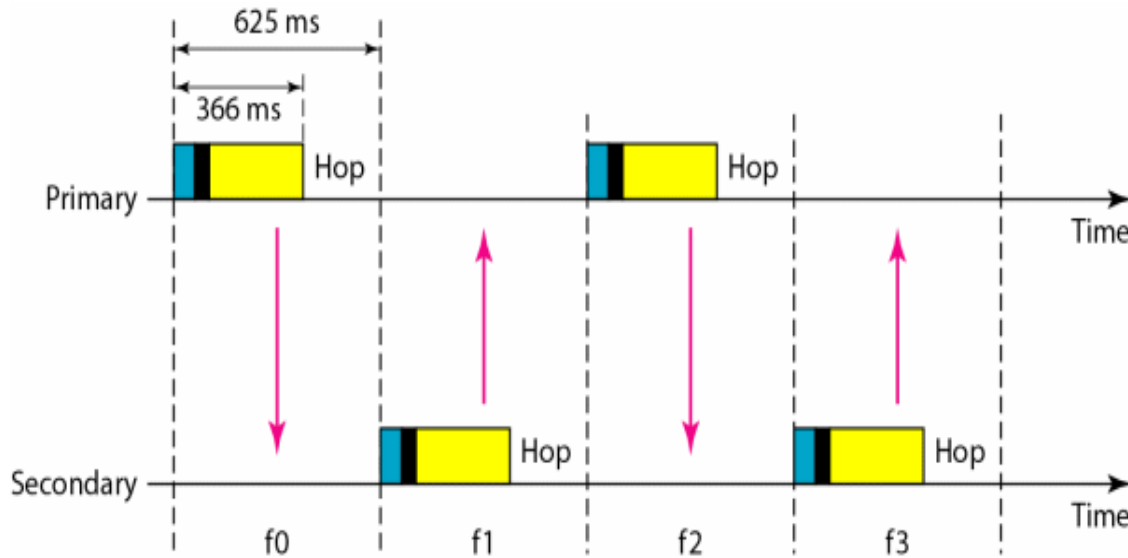


- Equivalent to the MAC sublayer
- Uses Time-Division Duplex TDMA
- Communication is between primary and secondaries only
- If there is one secondary, time slot is divided into 625 micro seconds with the primary using even-numbered slots and secondary using odd-numbered slots

Baseband Layer



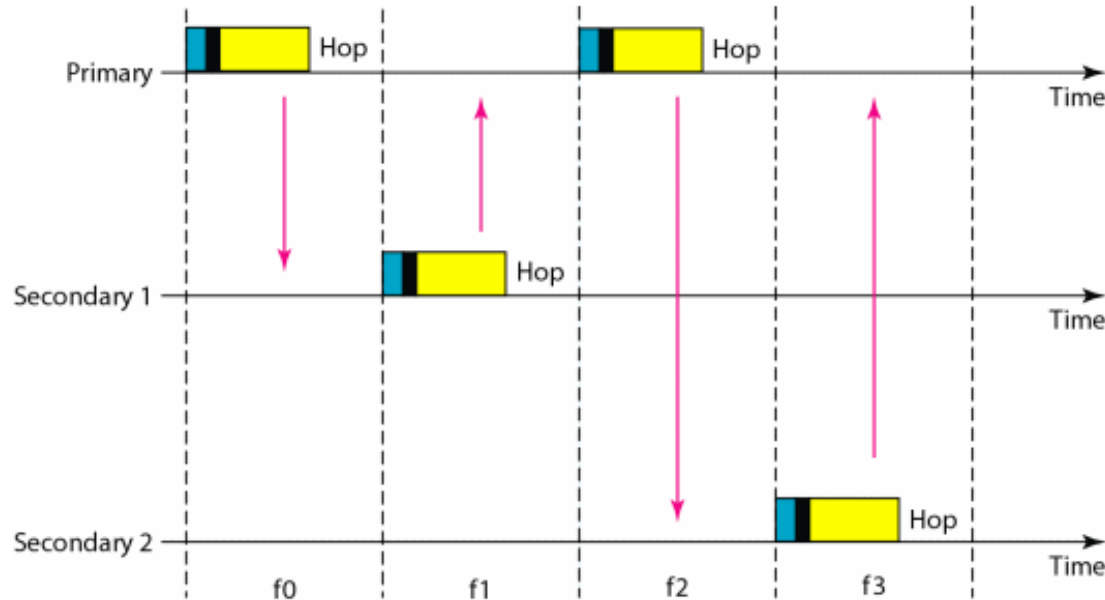
Single secondary communication; primary uses even numbered slots



Baseband Layer

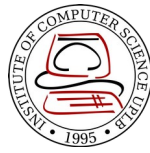


Multiple-secondary communication; similar to poll/select with reservations



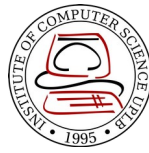
- **Synchronous connection-oriented** (SCO) – used when avoiding latency is more important than integrity; no re-transmission
 - Reserves two slots at regular intervals
 - Used in real-time audio
- **Asynchronous connectionless link** (ACL) – used when data integrity is more important than latency; with re-transmission

Frame Format

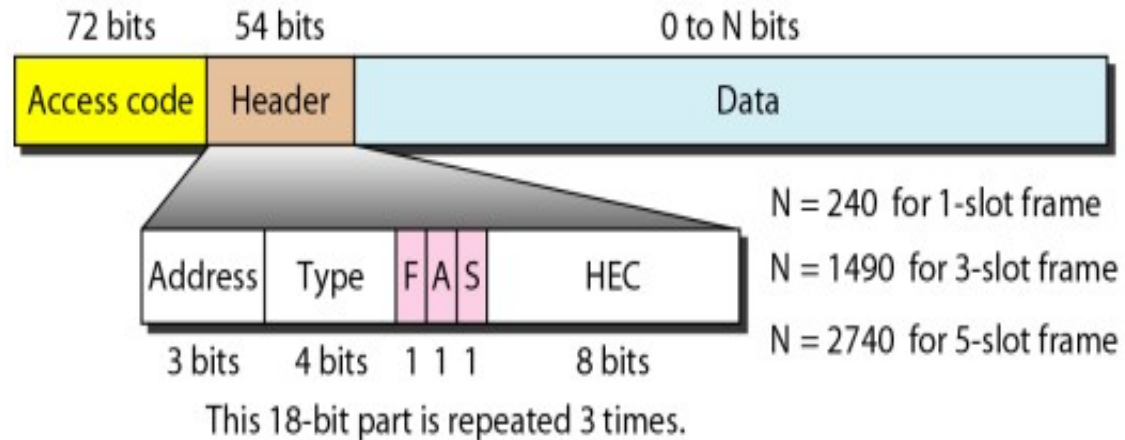


- Three types: one-slot, three-slot, five-slot
- A slot is 625 microseconds
- **One-slot**: with 1-MHz bandwidth and 1 bit/Hz, size is 366 bits
- **Three-slot**: size is 1616 bits
- **Five-slot**: size is 2866 bits

Frame Format



- **Access code** – 72-bits; sync bits; primary id
- **Header** – 54 bits in a repeated 18-bit pattern
 - Address – secondaries id
 - Type – type of data
 - F – flow control, A – ACK, S – seq #, HEC – checksum
- **Data** – 0 to 2740 bits



- Logical Link Control and Adaptation Protocol
- Roughly equivalent to the LLC sublayer
- Used for exchange in ACL link
- Allows multiplexing, segmentation and reassembly, QoS, and Group Management

References



- Electronic Notes. WiFi Standards: IEEE 802.11. <https://www.electronic-notes.com/articles/connectivity/wifi-ieee-802-11/standards.php>
- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.